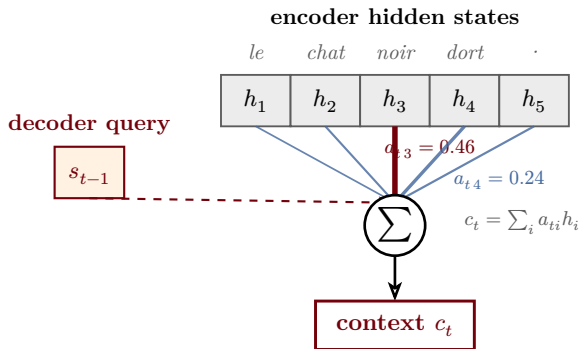


Bahdanau / Luong cross-attention

decoder query scores every encoder state $\cdot$ softmax $\rightarrow$ alignment weights $\cdot$ weighted-sum context vector



align weights: $a_{ti} = \text{softmax}_i(e_{ti})$

Bahdanau (additive): $e_{ti} = v^\top \tanh(W_s s_{t-1} + W_h h_i)$

Luong (mult.): $e_{ti} = s_{t-1}^\top h_i$ or $s_{t-1}^\top W h_i$

		alignment A_{ti}				
		le	chat	noir	dort	.
target steps t (English)	the	0.62	0.18	0.08	0.06	0.06
	black	0.10	0.20	0.58	0.07	0.05
	cat	0.07	0.11	0.46	0.24	0.12
	sleeps	0.05	0.07	0.12	0.70	0.06
	<eos>	0.06	0.06	0.06	0.14	0.68

source positions i (French)

The “cat” row of the heatmap (highlighted in panel A) shows the alignment weights for one decoder step; edge thickness $\propto$ weight.